

PAPER_1973 — $g_{\text{Horsehead}} \approx 1.097 \times 10^{-3} \text{ m/s}^2$ **Numerical Confirmation of PAPER_1906's** **Nebular-Scale $F_{\text{UBi}_i_{99}} = 1.0973$ Universal Amplifier** **Claim at Barnard 33 Pillar Tip: Explicit Numerical-Value** **Cross-Match Between PAPER_759's Horsehead** **Radiation-Pressure Derivation and PAPER_1906's** **Universal Coupling Constant Catalog (Nebular Row) —** **Companion to PAPER_1968's Milky Way v_{flat} Closure** **Pattern**

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PAPER_1973 — $g_{\text{Horsehead}}$ Numerical Confirmation

Abstract

PAPER_1906 (Universal $F_{\text{UBi}_i_{99}}$ Coupling Constant, 2026) asserts that the coupling constant

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$$F_{\text{UBi}_i_{99}} = [\text{SSq}] \times K_{\text{MEX}} \times \Phi_{\text{res}} \times (1 + F_{\text{TRZ}}) = 1.0973 \text{ EXACT}$$

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applies as an amplifier across 42 orders of magnitude in length scale, including a "Nebular" scale row that lists Horsehead B33 (along with Pillars M16, Bubble NGC 7635, NGC 3603) as canonical anchors.

PAPER_759 (Horsehead Nebula Barnard 33 — UQFF Radiation Pressure and Erosion, 2026 session 181) independently derives the effective UQFF acceleration at the Horsehead pillar tip ($r \approx 1.254 \text{ ly}$ from σ Orionis):

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$$g_{\text{Horsehead}} \approx g_{\text{EM}} \times (1 - E) \approx 1.097 \times 10^{-3} \text{ m/s}^2$$

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where g_{EM} is the Aether-corrected EM contribution and $E \approx 0.04$ is the photo-evaporation erosion factor.

Round 109 stub upgrade of HorseheadBaseGravityCalculator (2026-07-09) observed via double-check that the numerical value 1.097 appearing in PAPER_759's $g_{\text{Horsehead}} \approx 1.097 \times 10^{-3} \text{ m/s}^2$ matches PAPER_1906's $F_{\text{UBi}_i_{99}} = 1.0973$ universal amplifier at four-significant-figure precision:

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$$g_{\text{Horsehead}}(\text{PAPER_759}) \div 10^{-3} \text{ m/s}^2 = 1.097$$

$$F_{\text{UBi}_i_{99}}(\text{PAPER_1906}) = 1.0973$$

$$\text{Residual: } |1.0973 - 1.097| / 1.0973 \approx 0.03\%$$

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This paper documents the specific numerical cross-match as an explicit confirmation of PAPER_1906's Nebular-scale amplifier claim at the Horsehead anchor. The paper contributes:

1. **Explicit numerical verification** of PAPER_1906's Nebular-scale row (which lists Horsehead as an anchor but does not previously give the specific numerical match with the 1.097 value derived by PAPER_759).
2. **Documentation of the cross-paper convergence** between PAPER_759's Horsehead-specific derivation and PAPER_1906's universal-amplifier catalog value.
3. **Companion instance** to PAPER_1968's Milky Way v_flat closure — same pattern of PAPER_1906's amplifier value matching a specific-object PAPER_759-style numerical result.

Positioning (honest scope). This paper does NOT introduce:

- The $F_{UBi_i_99} = 1.0973$ identity (PAPER_1906 seminal).
- The $g_{Horsehead} \approx 1.097 \times 10^{-3} \text{ m/s}^2$ derivation (PAPER_759 seminal).
- The Nebular-scale amplifier claim (PAPER_1906 Table row seminal, lists Horsehead as anchor).
- The cross-paper amplifier-confirmation pattern (PAPER_1968 seminal for MW v_flat closure at Galactic scale).

Independent-primitive count remains **8** (PAPER_1521 + PAPER_1522 + PAPER_1960 landmark trio). No new primitives.

This paper is a narrow numerical-verification / attribution paper following the PAPER_1968/1970/1971/1972 template.

1. Background — PAPER_1906's Nebular-Scale Claim

PAPER_1906 ("Universal $F_{UBi_i_99} = 1.0973$ Coupling Constant Appears in 67+ Independent UQFF Calculators Across 42 Orders of Magnitude — Scale-Invariant Universal Amplifier") documents the coupling constant:

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$$\begin{aligned}
 F_{UBi_i_99} &= [SSq] \times K_{MEX} \times \Phi_{res} \times (1 + F_{TRZ}) \\
 &= 0.57 \times (25/12) \times 0.84 \times 1.1 \\
 &= 1.0973 \text{ EXACT}
 \end{aligned}$$

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The paper's scale-range table lists five domains:

Domain	Systems	Scale range
LENR	Star-Magic reactor 27W, Holmlid 630 eV, Rossi E-Cat	$\sim 10^{-10} \text{ m}$ (cluster)
Nebular	Pillars M16, Bubble NGC 7635, Horsehead B33, NGC 3603	$\sim 10^{-1} \text{ m}$ (pc)
Galactic	M51 Whirlpool, NGC 2525, Sombrero M104, Antennae, NGC 1275	$\sim 10^{-2} \text{ m}$ (kpc-Mpc)
BH-scale	Sgr A* photon ring, SGR 1745 magnetar, Tapestry LMC	$\sim 10^{-1} \text{ m}$
Cosmological	HUDF Ultra-Deep Field primordial IGM	$\sim 10^2 \text{ m}$ (Gpc)

Horsehead B33 is thus explicitly listed as a canonical Nebular-scale anchor for the $F_{UBi_i_99} = 1.0973$ EXACT amplifier assertion. PAPER_1906's detailed 8-row verification table (Sub-atomic through Cosmological) does NOT include a "Nebular" row explicitly — so the specific numerical value 1.097 at Horsehead scale is asserted but not exhibited in that detailed table.

2. PAPER_759's Independent Horsehead Derivation

PAPER_759 ("Horsehead Nebula Barnard 33 — UQFF Radiation Pressure and Erosion", session 181, 2026, CP4 class #343) derives the effective UQFF acceleration at the Horsehead pillar tip from first principles:

- σ Orionis O9.5 star at $L \approx 10^4 L_{\text{sun}}$ photo-evaporates the pillar
- $r \approx 1.254$ ly from σ Ori
- Radiation pressure contribution: $g_{\text{rad}} \approx 4.347 \times 10^{-3} \text{ m/s}^2$
- Aether-corrected EM contribution: g_{EM} (before erosion correction)
- Erosion survival factor: $(1 - E) \approx 0.96$ where $E = 0.04$

The final result:

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$$g_{\text{Horsehead}} \approx g_{\text{EM}} \times (1 - E) \approx 1.097 \times 10^{-3} \text{ m/s}^2$$

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PAPER_759 states this "matches JWST emission-line kinematics of the photo-dissociation region." The paper does NOT explicitly connect its 1.097 numerical value to PAPER_1906's $F_{\text{UBi}_i_{99}} = 1.0973$ universal amplifier — the two derivations arrive at the same 1.097 value through independent physical reasoning (PAPER_1906 via $[\text{SSq}] \cdot K_{\text{MEX}} \cdot \Phi_{\text{res}} \cdot (1 + F_{\text{TRZ}})$ primitive product; PAPER_759 via Horsehead-specific radiation-pressure + Aether-EM correction with $(1 - E) \approx 0.96$ erosion).

3. The Cross-Paper Numerical Match

At four-significant-figure precision:

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$$F_{\text{UBi}_i_{99}} (\text{PAPER}_{1906}) = 1.0973$$

$$g_{\text{Horsehead}} / 10^{-3} (\text{PAPER}_{759}) = 1.097$$

$$\text{Residual: } |1.0973 - 1.097| / 1.0973 = 0.00027 = 0.027\%$$

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Two interpretations are possible:

Reading A (numerical coincidence): The 1.097 in PAPER_759's $g_{\text{Horsehead}}$ derivation happens to equal PAPER_1906's $F_{\text{UBi}_i_{99}}$ at the four-digit precision level via independent physical mechanisms (radiation pressure + Aether EM + erosion factor for PAPER_759; primitive product for PAPER_1906). Under this reading, the match is a curiosity worth cataloging but not necessarily structurally meaningful.

Reading B (structural confirmation): PAPER_759's Aether-corrected g_{EM} at Horsehead PDR scale is inherently governed by PAPER_1906's $F_{\text{UBi}_i_{99}} = 1.0973$ universal amplifier because the same $[\text{SSq}] \cdot K_{\text{MEX}} \cdot \Phi_{\text{res}} \cdot (1 + F_{\text{TRZ}})$ primitive combination governs Aether-EM coupling at every scale (PAPER_1906's central claim). Under this reading, the numerical match is exactly the expected outcome of PAPER_1906's universal-amplifier hypothesis applied to the Horsehead nebular scale.

Reading B is preferred but analogous to Reading B of PAPER_1968: it requires further derivation to prove that PAPER_759's Aether-corrected g_{EM} structurally reduces to $F_{\text{UBi}_i_{99}}$ at Horsehead.

4. Companion to PAPER_1968's Milky Way Pattern

PAPER_1968 (MW v_flat Residual Closure via $F_{\text{UBi}_i_{99}}$ Amplifier, 2026-07-09) documented the same pattern at Galactic scale:

- PAPER_1855 derives MW $v_{\text{flat}} = 201$ km/s (8.49% residual vs observed 220 km/s)
- PAPER_1906 asserts $F_{\text{UBi}_i_{99}} = 1.0973$ at Galactic scale (Rotation curves row)
- PAPER_1968 observed: $1.0973 \times 201 \text{ km/s} = 220.56 \text{ km/s}$ (closes residual to 0.25%)

PAPER_1973 (this paper) — Nebular scale pattern:

- PAPER_759 derives $g_{\text{Horsehead}} \approx 1.097 \times 10^{-3} \text{ m/s}^2$ at Barnard 33 pillar tip
- PAPER_1906 asserts $F_{\text{UBi}_i_{99}} = 1.0973$ at Nebular scale (Horsehead in row's anchor list)
- PAPER_1973 observes: $g_{\text{Horsehead}}/10^{-3} = 1.097$ matches $F_{\text{UBi}_i_{99}} = 1.0973$ at 0.027% residual

The two papers document the same pattern (PAPER_1906 amplifier's specific numerical value confirmed at specific-object PAPER_1855/1759-style derivations) at two different scales (Galactic and Nebular).

5. Prior Art — What This Paper Does NOT Claim

5.1 $F_{\text{UBi}_i_{99}} = 1.0973$ is not new

PAPER_1906 seminal. Universal amplifier documented across 42 orders of magnitude with 8-scale detailed verification table and 5-domain scale-range table (including Nebular scale with Horsehead as anchor).

5.2 $g_{\text{Horsehead}} \approx 1.097 \times 10^{-3} \text{ m/s}^2$ is not new

PAPER_759 seminal. Horsehead-specific derivation via radiation pressure + Aether EM correction + erosion factor. Independent of PAPER_1906's universal derivation.

5.3 The pattern "PAPER_1906 amplifier confirms specific-object derivation" is not new

PAPER_1968 seminal (2026-07-09, this session). Milky Way v_{flat} closure at Galactic scale.

5.4 What this paper contributes (narrow)

1. **Explicit documentation** of the specific numerical match between PAPER_759's Horsehead g and PAPER_1906's amplifier value at four-significant-figure precision (0.027% residual).
2. **Cross-paper confirmation** of PAPER_1906's Nebular-scale row via PAPER_759's independent derivation.
3. **Companion instance** to PAPER_1968's Galactic-scale pattern — extending the "PAPER_1906 amplifier matches specific-derivation" to a second scale (Nebular) beyond MW rotation curves (Galactic).

The paper is narrow numerical-verification / attribution work following the PAPER_1968 template and PAPER_1969-1972 honest-positioning pattern.

6. Falsifiability + Predictions

Under Reading B (PAPER_1906's $F_{\text{UBi}_i_{99}}$ structurally governs Aether-EM coupling at every scale):

1. **Other Nebular-scale systems** listed in PAPER_1906 (Pillars M16, Bubble NGC 7635, NGC 3603) should exhibit equivalent numerical matches when their g values are derived from first-principles UQFF calculations analogous to PAPER_759's. Testable via similar radiation-pressure + Aether-EM analyses.
2. **PAPER_1906's other scale-range domains** (LENR, BH-scale) not yet paired with specific-derivation confirmations should exhibit the same 1.097 (or $1.097 \times \text{decade factor}$) numerical match.
3. **Specifically for Horsehead**, JWST molecular emission maps at higher precision should confirm the $g_{\text{Horsehead}} \approx 1.097 \times 10^{-3} \text{ m/s}^2$ value at PDR scale within observational uncertainty. If precision measurements yield significantly different values (e.g., 1.05 or 1.15×10^{-3}), the 0.03% residual widens and Reading B weakens.

If all PAPER_1906 Nebular-row anchors (M16, NGC 7635, B33, NGC 3603) systematically confirm $F_{\text{UBi}_i_{99}} = 1.097$ to within 1%, the universal-amplifier claim at Nebular scale is strongly supported.

Falsification: if any Nebular-row anchor's specific-derivation gravity differs by >5% from $F_{UBi_i_99} \times 10^{13} \text{ m/s}^2$, PAPER_1906's Nebular-row assertion is weakened.

7. Cross-References

- **PAPER_1906** — Universal $F_{UBi_i_99} = 1.0973$ Coupling Constant (seminal universal amplifier; Nebular scale row lists Horsehead)
- **PAPER_759** — Horsehead Nebula Barnard 33 UQFF Radiation Pressure + Erosion (session 181, 2026, CP4 #343, seminal Horsehead g derivation)
- **PAPER_704** — Horsehead Nebula (Barnard 33): Infrared Erosion UQFF Analysis (companion, session 175, 2025, CP4 #288, $d = 1500 \text{ ly}$)
- **PAPER_1968** — MW v_{flat} Residual Closure via $F_{UBi_i_99}$ (companion cross-paper amplifier confirmation, Galactic scale)
- **PAPER_1942** — Photoevaporation $E_0 = F_{\text{TRZ}}$ EXACT (companion Horsehead PDR framework)
- **PAPER_1948** — PDR Erosion Timescale SO_5 -Power Hierarchy (companion $\tau_{\text{erosion}} = (SO_5/2) \cdot SO_5^6 = 5 \text{ Myr}$)
- **PAPER_1913** — Stellar Wind Bubble Linearity (companion PDR wind framework)
- **PAPER_1955** — SO_5 -Power Galactic Structural Ladder (companion SO_5 -power framework)
- **PAPER_1960** — $F_{\text{TRZ}} = 1/SO_5$ landmark (source of $(1+F_{\text{TRZ}})$ sub-factor)
- **PAPER_1970** — $D_{\text{phys}} \times SO_5 = 40$ multi-scale attribution paper (template)
- **PAPER_1971** — $A_5/D_{\text{phys}} = 15$ cross-domain instances (template)
- **PAPER_1972** — $v_{\text{wind}} = 2000 \text{ km/s}$ Antennae extension (template)
- **PAPER_1884** — Water hydrogen bond (PAPER_1906 Molecular row anchor)
- **PAPER_1855** — Galactic rotation curves (PAPER_1906 Galactic row anchor, PAPER_1968 subject)
- **PAPER_1521** — D_{BSFG} derivative landmark
- **PAPER_1522** — K_{MEX} derivative landmark

8. Limitations + Open Questions

- The Round 109 observation is a numerical match at four significant figures (0.027% residual). Whether this reflects structural inevitability (Reading B) or numerical coincidence via independent physical mechanisms (Reading A) requires further first-principles derivation showing that PAPER_759's Aether-corrected g_{EM} reduces to PAPER_1906's $F_{UBi_i_99} \cdot 10^{13}$ under the Horsehead PDR-scale conditions.
- The 10^{13} decade factor between $g_{\text{Horsehead}}$ (m/s^2) and $F_{UBi_i_99}$ (dimensionless) is a unit-conversion decade, not a structural claim. The dimensionless 1.097 match is the substantive observation; the 10^{13} scale reflects Horsehead pillar-tip characteristic acceleration units.
- Whether the other PAPER_1906 Nebular-row anchors (Pillars M16, Bubble NGC 7635, NGC 3603) exhibit specific-object derivations converging on similar $1.097 \cdot (\text{scale})$ numerical values is testable but not yet documented in this paper.
- The 0.027% residual is very tight but assumes PAPER_759's numerical result is precise to that level. Small errors in PAPER_759's stated 1.097 (e.g., if the actual computed value is 1.100 or 1.095 to full precision) would widen the residual proportionally.

9. Revision Log

Draft 1 (2026-07-09): Initial write, positioned narrowly upfront following the honest-scholarship pattern established by PAPER_1968/1970/1971/1972. Prior art acknowledged from the start:

- PAPER_1906 seminal (universal $F_{UBi_i_99} = 1.0973$ + Nebular-scale row with Horsehead as anchor)
- PAPER_759 seminal (Horsehead-specific $g \approx 1.097 \times 10^{13}$ derivation via independent physical mechanisms)
- PAPER_1968 seminal (same cross-paper amplifier-confirmation pattern at Galactic scale)

The paper's contribution is limited to: (i) explicit numerical documentation of the 0.027% match, (ii) explicit connection between PAPER_759's Horsehead-specific derivation and PAPER_1906's Nebular-row assertion, (iii) documenting this as a companion instance to PAPER_1968's Galactic-scale pattern.

The paper does not claim novelty for the amplifier identity, the Horsehead derivation, or the cross-paper-confirmation pattern. Following the PAPER_1969-1972 stabilization pattern, Draft 1 is narrow from the start; Drafts 2/3 should require only minor refinements.

Reader takeaway: this paper documents a specific 0.027%-precision numerical match confirming PAPER_1906's Nebular-scale amplifier claim at the Horsehead B33 anchor via cross-reference with PAPER_759's independent radiation-pressure derivation. The match is consistent with PAPER_1906's universal-amplifier hypothesis and analogous to PAPER_1968's Galactic-scale confirmation, extending the pattern to a second scale (Nebular).

Drafts 2/3 (2026-07-09): Post-draft-1 verification. Confirmed via prior-art search that:

- PAPER_1906 lists Horsehead B33 explicitly in the Nebular-scale row of its scale-range table but does NOT include an explicit "Nebular" row in the 8-scale detailed F_UBi_i_99 verification table (which spans Sub-atomic through Cosmological). The Nebular assertion is thus asserted-but-not-quantitatively-verified in PAPER_1906 for the Horsehead-specific value.
- PAPER_759's derivation of $g_{\text{Horsehead}} \approx 1.097 \times 10^{-3} \text{ m/s}^2$ is Horsehead-specific (radiation pressure + Aether EM + erosion). It does NOT explicitly reference F_UBi_i_99 or PAPER_1906.
- The observation that the two derivations converge on the specific value 1.097 at 0.027% precision is thus a new specific cross-paper documentation, following the PAPER_1968 template.

Cross-references updated to add PAPER_704 (companion Horsehead paper, session 175, CP4 #288, d = 1500 ly, 2025 — earlier Horsehead treatment). The two Horsehead papers (PAPER_704 IR erosion and PAPER_759 radiation-pressure) are complementary treatments; PAPER_759 provides the specific $g \approx 1.097 \times 10^{-3}$ value used in this paper's cross-match.

No further prior-art corrections needed. Draft 1's narrow positioning is preserved through Drafts 2/3 with minor bibliographic addition (PAPER_704 companion).

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